

Machine Learning Slot-2

12 Practice Papers with Detailed Answers

Exam-style question bank: concepts, numericals, edge cases, limitations, and interpretations

What this PDF covers	Decision trees, rule-based classifiers, KNN, k-means, DBSCAN, BIRCH, association rules, regression, ensembles, evaluation metrics, preprocessing, PCA, SVM, Naive Bayes, and edge cases.
Answer style	Every answer follows: definition -> formula/steps -> calculation -> interpretation -> exam-ready wording.
Best use	First solve each paper yourself, then check the answer guide. Pay extra attention to formulas and edge cases.

High-yield pattern: your exam is testing not only definitions, but also what happens when an algorithm faces unusual conditions: empty clusters, k too small/large, varying density, data leakage, high confidence but low lift, and outlier sensitivity.

Quick Formula Sheet

Topic	Formula / Key idea
Entropy	$\text{Entropy}(D) = - \sum p_i \log_2(p_i)$
Information Gain	$\text{IG} = \text{Entropy}(\text{parent}) - \text{Weighted Entropy}(\text{children})$
SplitInfo	$\text{SplitInfo} = - \sum (N_j / P) \log_2(N_j / P)$
Gain Ratio	$\text{GainRatio} = \text{InformationGain} / \text{SplitInfo}$
Gini	$\text{Gini}(D) = 1 - \sum p_i^2$
Misclassification Error	$\text{ME} = 1 - \max(p_i)$
Rule Coverage	$\text{Coverage} = \text{records covered by rule} / \text{total records}$
Rule Accuracy	$\text{Accuracy} = \text{correctly classified covered records} / \text{covered records}$
Euclidean Distance	$d(x,y) = \sqrt{\sum (x_j - y_j)^2}$
K-means SSE	$\text{SSE} = \sum \text{over clusters} \sum \text{over points} \ x - \text{centroid}\ ^2$
Min-Max Normalization	$x' = (x - \min) / (\max - \min)$
Z-score	$z = (x - \text{mean}) / \text{standard deviation}$
MSE / RMSE / MAE	MSE = average squared error, RMSE = $\sqrt{\text{MSE}}$, MAE = average absolute error
Association Support	$\text{Support}(X) = \text{transactions containing } X / \text{total transactions}$
Confidence	$\text{Confidence}(X \rightarrow Y) = \text{Support}(X \cup Y) / \text{Support}(X)$
Lift	$\text{Lift}(X \rightarrow Y) = \text{Confidence}(X \rightarrow Y) / \text{Support}(Y)$
Possible itemsets	With I unique items: total non-empty itemsets = $2^I - 1$
Possible rules	With I unique items: total rules = $3^I - 2^{(I+1)} + 1$
Classification metrics	Accuracy, Precision, Recall, F1, Specificity from TP, FP, FN, TN

Practice Paper 1: Preprocessing, KNN, k-means, Random Forest

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain data preprocessing. Why are missing values, noisy data, and outliers dangerous for ML models? B) Given values 10,20,30,40,50, perform min-max normalization to [0,1], z-score normalization with mean=30 and standard deviation=14.14, and decimal scaling.

Answer Guide

Data preprocessing converts raw data into clean, consistent, model-ready data. It includes cleaning, integration, transformation, normalization, binning, reduction, and handling missing values/outliers.

- Missing values can make formulas invalid or bias the model if filled badly.
- Noisy data adds random error and can teach the model wrong patterns.
- Outliers can dominate distance, mean, SSE, and regression coefficients.

Numerical solution

Data = 10,20,30,40,50; min=10; max=50; mean=30; sigma=14.14

Min-max: $x' = (x - \min) / (\max - \min) = (x - 10) / 40$

x	Min-max
10	0
20	0.25
30	0.50
40	0.75
50	1.00

Z-score: $z = (x - \text{mean}) / \text{sigma} = (x - 30) / 14.14$

x	z-score
10	-1.41
20	-0.71
30	0
40	0.71
50	1.41

Decimal scaling: divide by 100 because max absolute value is 50, so $j=2$. Values: 0.10, 0.20, 0.30, 0.40, 0.50

Interpretation: normalization preserves order but changes scale. It is especially important for distance-based algorithms like KNN and k-means.

Q2. A) Explain KNN as a lazy learner. Why is feature scaling important in KNN? B) Classify $X=(3,3)$ using $k=3$ and Euclidean distance. Training: A(1,2) Red, B(2,3) Red, C(6,6) Blue, D(7,7) Blue, E(3,2) Red.

Answer Guide

KNN is a lazy learner because it does not build a strong explicit model during training. It stores the training data and performs most computation during prediction.

Scaling matters because KNN uses distance. A feature with a large numeric range can dominate the distance and make other features almost irrelevant.

Euclidean distance: $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Point	Calculation	Distance	Class
A(1,2)	$\sqrt{(3-1)^2+(3-2)^2}$	$\sqrt{5}=2.24$	Red
B(2,3)	$\sqrt{(3-2)^2+(3-3)^2}$	1.00	Red
C(6,6)	$\sqrt{(3-6)^2+(3-6)^2}$	4.24	Blue
D(7,7)	$\sqrt{(3-7)^2+(3-7)^2}$	5.66	Blue
E(3,2)	$\sqrt{(3-3)^2+(3-2)^2}$	1.00	Red

The 3 nearest neighbors are B, E, and A. Their labels are Red, Red, Red.

Final prediction = Red

Q3. A) Explain why k-means fails for non-spherical clusters such as concentric circles. B) Points are 1,2,3,10,11,12. Let $k=2$ and initial centroids $C1=1$, $C2=10$. Perform one iteration and calculate new centroids.

Answer Guide

K-means assigns points to the nearest centroid and minimizes SSE. This naturally prefers compact, spherical/blob-like clusters. For rings, moons, spirals, or concentric circles, the true cluster may wrap around another cluster, but centroid distance cuts space into convex regions.

One k-means iteration

Point	Distance to $C1=1$	Distance to $C2=10$	Assigned cluster
1	0	9	C1
2	1	8	C1
3	2	7	C1
10	9	0	C2
11	10	1	C2
12	11	2	C2

New $C1 = \text{mean}(1,2,3)=2$; New $C2 = \text{mean}(10,11,12)=11$

Interpretation: k-means repeats assignment and centroid update until assignments stop changing.

Q4. A) Explain Random Forest and why it improves over a single decision tree. B) Explain five evaluation metrics: accuracy, precision, recall, F1-score, specificity.

Answer Guide

Random Forest is an ensemble of decision trees. Each tree is trained on a bootstrap sample of rows, and at each split it considers only a random subset of features. Final classification uses majority voting; regression uses averaging.

- Single decision tree: high variance, can overfit, sensitive to small data changes.
- Random Forest: many diverse trees; voting reduces variance and improves generalization.
- OOB samples can provide internal validation because each tree leaves out some rows.

$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$ -> among predicted positives, how many are truly positive

$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$ -> among actual positives, how many were detected

$\text{F1} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$

$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$ -> among actual negatives, how many were correctly rejected

Practice Paper 2: Decision Trees, Rules, DBSCAN, AdaBoost

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain Decision Tree induction and deduction. B) Parent node has 8X,2Y. Split: N1=5X,0Y and N2=3X,2Y. Calculate parent entropy, child entropies, weighted entropy, and information gain.

Answer Guide

Tree induction is the training process: recursively choose attributes/splits that reduce impurity. Tree deduction is prediction: pass a new record from root to leaf and output the leaf class.

$$\text{Entropy}(D) = - \sum p_i \log_2(p_i)$$

$$\text{Parent: } pX=8/10=0.8, \quad pY=2/10=0.2$$

$$\text{Entropy}(P) = -(0.8 \log_2(0.8) + 0.2 \log_2(0.2)) = 0.722$$

$$N1: 5X, 0Y \rightarrow \text{Entropy}(N1)=0$$

$$N2: 3X, 2Y \rightarrow pX=3/5=0.6, \quad pY=2/5=0.4 \rightarrow \text{Entropy}(N2)=0.971$$

$$\text{Weighted entropy} = (5/10)*0 + (5/10)*0.971 = 0.486$$

$$\text{Information Gain} = 0.722 - 0.486 = 0.236$$

The split is useful because impurity decreases from 0.722 to 0.486.

Q2. A) Explain rule-based classifier and properties of good rules. B) Dataset has 200 records. A rule covers 50 records and 40 are correctly classified. Calculate coverage and accuracy; interpret.

Answer Guide

A rule-based classifier uses IF-THEN rules. The IF part is the antecedent; the THEN part is the consequent/class prediction.

- Good rules should be accurate, have sufficient coverage, be interpretable, be consistent, avoid conflict, and be simple.
- A complete rule set should cover most or all records; a default rule may be used for uncovered cases.

$$\text{Coverage} = \text{records covered} / \text{total records} = 50/200 = 0.25 = 25\%$$

$$\text{Rule accuracy} = \text{correct covered} / \text{covered} = 40/50 = 0.80 = 80\%$$

The rule is reasonably accurate and has moderate coverage. It is useful, but we must compare it with other rules and check whether it overfits.

Q3. A) Explain DBSCAN core, border, and noise points. B) DBSCAN uses epsilon=2, MinPts=3. For point P, there are 2 other points within epsilon. If counting P itself, decide whether P is core, border, or noise. Explain.

Answer Guide

- Core point: has at least MinPts points in its epsilon-neighborhood, usually including itself.
- Border point: not core, but lies within epsilon of a core point.
- Noise point: neither core nor border.

Here P has 2 other points within epsilon. Counting P itself gives neighborhood size = 3.

$$\text{Neighborhood size} = 2 + 1 = 3; \quad \text{MinPts} = 3$$

Since $3 \geq 3$, P is a core point.

Exam caution: Some definitions count the point itself; most DBSCAN explanations do. State your assumption clearly.

Q4. A) Explain bagging vs boosting. B) In AdaBoost, initially 5 samples have equal weight. A weak learner misclassifies 2 samples. Calculate weighted error and explain what happens to misclassified

sample weights.**Answer Guide**

Aspect	Bagging	Boosting
Training	Parallel/independent	Sequential
Data focus	Bootstrap samples	Misclassified samples get more weight
Main effect	Reduces variance	Often reduces bias
Example	Random Forest	AdaBoost

Initial weight per sample = $1/5 = 0.2$

Misclassified samples = 2 -> weighted error = $2 * 0.2 = 0.4$

AdaBoost increases the weights of misclassified samples and decreases the relative weights of correctly classified samples. The next weak learner focuses more on the hard examples.

Practice Paper 3: Data Types, Gini, Elbow Method, Association Rules

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain nominal, ordinal, interval, and ratio attributes with examples. B) Identify the attribute type: blood group, education level low/medium/high, temperature in Celsius, salary, age.

Answer Guide

Attribute type	Meaning	Example
Nominal	Labels with no order	blood group, city
Ordinal	Ordered categories but unequal/unknown gaps	low/medium/high
Interval	Meaningful difference, no true zero	Celsius temperature
Ratio	Meaningful difference and true zero	salary, age, weight

Feature	Type
Blood group	Nominal
Education level low/medium/high	Ordinal
Temperature in Celsius	Interval
Salary	Ratio
Age	Ratio

Q2. A) Explain Gini Index and why it is used in decision trees. B) Parent node 6A,4B. Split: N1=4A,0B and N2=2A,4B. Calculate Gini of parent, N1, N2, weighted Gini, and Gini reduction.

Answer Guide

Gini Index measures node impurity. It is used in CART-style decision trees to choose splits that make child nodes purer.

$$\text{Gini}(D) = 1 - \sum p_i^2$$

$$\text{Parent: } p_A=0.6, p_B=0.4 \rightarrow \text{Gini}(P) = 1 - (0.36 + 0.16) = 0.48$$

$$\text{N1: } p_A=1, p_B=0 \rightarrow \text{Gini}(N1) = 0$$

$$\text{N2: } p_A=2/6=0.333, p_B=4/6=0.667 \rightarrow \text{Gini}(N2) = 1 - (0.333^2 + 0.667^2) = 0.444$$

$$\text{Weighted Gini} = (4/10) * 0 + (6/10) * 0.444 = 0.267$$

$$\text{Gini Reduction} = 0.48 - 0.267 = 0.213$$

Q3. A) Explain the elbow method for choosing k in k-means. B) Given SSE values k=1:500, k=2:250, k=3:120, k=4:100, k=5:92. Which k would you choose and why?

Answer Guide

The elbow method plots k against SSE/inertia. SSE always decreases as k increases, so we do not simply choose the smallest SSE. We choose the point where improvement starts becoming small.

k	SSE	Drop from previous
1	500	-
2	250	250
3	120	130
4	100	20
5	92	8

The large drops occur until $k=3$. After $k=3$, improvement is small.

Choose $k=3$

Interpretation: $k=3$ is the elbow point because it balances compactness with model simplicity.

Q4. A) Explain association rule mining. Define support, confidence, and lift. B) In 100 transactions: 40 contain Milk, 30 contain Bread, 20 contain both. Calculate support of {Milk,Bread}, confidence of Milk -> Bread, and lift.

Answer Guide

Association rule mining finds patterns of co-occurrence in transaction data, such as Milk -> Bread.

$\text{Support}(X) = \text{transactions containing } X / \text{total transactions}$

$\text{Confidence}(X \rightarrow Y) = \text{Support}(X \text{ union } Y) / \text{Support}(X)$

$\text{Lift} = \text{Confidence}(X \rightarrow Y) / \text{Support}(Y)$

$\text{Support}(\text{Milk, Bread}) = 20/100 = 0.20$

$\text{Confidence}(\text{Milk} \rightarrow \text{Bread}) = 20/40 = 0.50$

$\text{Support}(\text{Bread}) = 30/100 = 0.30$

$\text{Lift} = 0.50/0.30 = 1.67$

Lift > 1 means Milk and Bread have a positive association. Milk buyers are 1.67 times more likely to buy Bread compared with a random transaction.

Practice Paper 4: Overfitting, Gain Ratio, Hierarchical, Naive Bayes

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain overfitting, underfitting, and good fit using training and test error. B) Training accuracy=99%, test accuracy=65%. Identify the problem and suggest two remedies.

Answer Guide

- Underfitting: model too simple; high training error and high test error.
- Overfitting: model memorizes training data; low training error but high test error.
- Good fit: training and test performance are both reasonably good and close.

Training accuracy 99% and test accuracy 65% indicates overfitting.

- Remedies: pruning for trees, regularization, simpler model, more data, cross-validation, feature selection, early stopping, or data augmentation depending on model type.

Q2. A) Explain SplitInfo and Gain Ratio. Why is Gain Ratio used instead of only Information Gain? B) Parent has 12 records. Split creates N1=3, N2=3, N3=6. Calculate SplitInfo.

Answer Guide

SplitInfo measures how data is distributed across branches. Gain Ratio normalizes Information Gain by SplitInfo, reducing the bias of Information Gain toward attributes with many branches.

$$\text{SplitInfo} = - \sum (|N_j|/|P|) \log_2(|N_j|/|P|)$$

Proportions: $3/12=0.25$, $3/12=0.25$, $6/12=0.50$

$$\text{SplitInfo} = -(0.25 \log_2(0.25) + 0.25 \log_2(0.25) + 0.5 \log_2(0.5))$$

$$= -[0.25(-2) + 0.25(-2) + 0.5(-1)] = -[-0.5 - 0.5 - 0.5] = 1.5$$

SplitInfo = 1.5

Q3. A) Explain hierarchical agglomerative clustering. B) Points are 0,4,5,20. Using single linkage, show the merge order and merge distances.

Answer Guide

Agglomerative clustering starts with each point as its own cluster and repeatedly merges the closest clusters until one cluster remains or desired number is reached.

Single linkage defines distance between clusters as the minimum pairwise distance.

Initial distances: $d(4,5)=1$, $d(0,4)=4$, $d(0,5)=5$, $d(5,20)=15$, $d(4,20)=16$, $d(0,20)=20$

1. Merge 4 and 5 at distance 1 -> cluster {4,5}.
2. Distance {4,5} to 0 = $\min(4,5)=4$. Distance {4,5} to 20 = $\min(16,15)=15$.
3. Merge 0 with {4,5} at distance 4 -> cluster {0,4,5}.
4. Distance {0,4,5} to 20 = $\min(20,16,15)=15$.
5. Merge {0,4,5} with 20 at distance 15.

Single linkage can create chaining because it connects clusters through nearest points.

Q4. A) Explain Naive Bayes and its independence assumption. B) Why does Naive Bayes often perform well in text classification even though the independence assumption is unrealistic?

Answer Guide

Naive Bayes is a probabilistic supervised classifier based on Bayes' theorem. It predicts the class with the highest posterior probability.

$$P(C|X) \text{ proportional to } P(C) * \text{product } P(x_j|C)$$

The 'naive' assumption is that features are conditionally independent given the class.

In text classification, words are not truly independent, but Naive Bayes often works well because word frequencies still provide strong class evidence. It is fast, handles high-dimensional sparse data, and gives strong baselines for spam filtering and document classification.

Practice Paper 5: Regression, Least Squares, Gradient Descent, Regularization

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain classification vs regression. B) Actual $y=[3,5,7,9]$, predicted $\hat{y}=[2,5,8,10]$. Calculate SSE, MSE, RMSE, and MAE.

Answer Guide

Classification predicts discrete class labels, such as spam/not spam. Regression predicts continuous values, such as price, marks, or demand.

y	yhat	error=y-yhat	squared error	absolute error
3	2	1	1	1
5	5	0	0	0
7	8	-1	1	1
9	10	-1	1	1

$$SSE=1+0+1+1=3$$

$$MSE=SSE/n=3/4=0.75$$

$$RMSE=\sqrt{0.75}=0.866$$

$$MAE=(1+0+1+1)/4=0.75$$

Q2. A) Explain the least squares method in linear regression. Why does it minimize SSE? B) Given data (1,2), (2,4), (3,5), find the least-squares line using slope/intercept formulas.

Answer Guide

Least squares fits the line $\hat{y}=mx+c$ that minimizes the sum of squared vertical errors between actual and predicted values.

x	y	x^2	xy
1	2	1	2
2	4	4	8
3	5	9	15
Sum	11? wait	14	25

Correct sums: $n=3$, $\sum x=6$, $\sum y=11$, $\sum x^2=14$, $\sum xy=25$.

$$m = [n \cdot \sum(xy) - \sum(x) \sum(y)] / [n \cdot \sum(x^2) - (\sum x)^2]$$

$$m = [3 \cdot 25 - 6 \cdot 11] / [3 \cdot 14 - 6^2] = (75 - 66) / (42 - 36) = 9/6 = 1.5$$

$$c = [\sum y - m \sum x] / n = [11 - 1.5 \cdot 6] / 3 = (11 - 9) / 3 = 0.667$$

$$\text{Least-squares line: } \hat{y} = 1.5x + 0.667$$

Q3. A) Explain why gradient descent is still needed when least squares gives a direct solution. B) What happens if learning rate is too small, too large, or properly chosen?

Answer Guide

Least squares gives a closed-form solution for ordinary linear regression, but it requires matrix operations/inversion in multiple dimensions. Gradient descent is more scalable and works for models/losses with no simple closed form, such as logistic regression and neural networks.

- Too small learning rate: very slow convergence.

- Too large learning rate: overshoots minimum, may diverge.
- Proper learning rate: cost decreases steadily toward a minimum.

Update idea: $\text{parameter} := \text{parameter} - \text{learning_rate} * \text{gradient}$

Q4. A) Explain L1 and L2 regularization. B) Which regularization can help feature selection and why?

Answer Guide

Regularization adds a penalty to the loss function to reduce overfitting by discouraging overly large weights.

L1 penalty = $\lambda * \sum |w_j|$

L2 penalty = $\lambda * \sum w_j^2$

L1 can push some weights exactly to zero, so it can perform feature selection. L2 usually shrinks weights but rarely makes them exactly zero.

Practice Paper 6: Binning, Leakage, k-means SSE, BIRCH

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain equal-width and equal-frequency binning. B) Sorted values: 5,7,8,10,12,14,18,21,25,30. Partition into 3 bins using equal-frequency and equal-width.

Answer Guide

Equal-frequency binning creates bins with roughly equal numbers of records. Equal-width binning creates numeric intervals with equal range width.

Equal-frequency with 10 values and 3 bins cannot be perfectly equal. One common split is 4-3-3.

Equal-frequency bins: $B1=\{5, 7, 8, 10\}$, $B2=\{12, 14, 18\}$, $B3=\{21, 25, 30\}$

Equal-width: $\min=5$, $\max=30$, $\text{width}=(30-5)/3=8.33$

Intervals: $[5, 13.33)$, $[13.33, 21.67)$, $[21.67, 30]$

Equal-width bins: $B1=\{5, 7, 8, 10, 12\}$, $B2=\{14, 18, 21\}$, $B3=\{25, 30\}$

Q2. A) Explain data leakage with examples. B) Why is normalizing before train-test split wrong?

Answer Guide

Data leakage happens when information unavailable at training/prediction time enters the training process, making performance look better than reality.

Normalizing before train-test split is wrong because min, max, mean, or standard deviation from the test set influence the training transformation.

- Correct process: split first, fit scaler on training data only, transform training and test using training scaler.

- Leakage causes overly optimistic evaluation because the model indirectly sees test-set statistics.

Q3. A) Explain k-means objective function and SSE. B) Clusters: $C1=\{1,2,3\}$, $C2=\{10,11,12\}$. Calculate centroid of each cluster, SSE of each cluster, and total SSE.

Answer Guide

K-means minimizes within-cluster SSE: total squared distance between each point and its assigned cluster centroid.

Centroid $C1=(1+2+3)/3=2$

$SSE1=(1-2)^2+(2-2)^2+(3-2)^2=1+0+1=2$

Centroid $C2=(10+11+12)/3=11$

$SSE2=(10-11)^2+(11-11)^2+(12-11)^2=1+0+1=2$

Total $SSE=2+2=4$

Q4. A) Explain BIRCH and why it is scalable. B) For points (1,2), (2,1), (2,2), calculate N, LS, SS, and centroid.

Answer Guide

BIRCH is a scalable clustering method that summarizes data into Clustering Features stored in a CF-tree. It is efficient because it compresses many raw points into micro-cluster summaries.

$CF=(N, LS, SS)$

$N=3$

$LS=(\sum x, \sum y)=(1+2+2, 2+1+2)=(5, 5)$

$SS=(1^2+2^2)+(2^2+1^2)+(2^2+2^2)=5+5+8=18$

Centroid= $LS/N=(5/3, 5/3)=(1.67, 1.67)$

Practice Paper 7: Bias-Variance, KNN Edges, SVM, Apriori

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain bias-variance tradeoff. B) Match cases: model too simple, model too complex, $k=1$ in KNN, $k=n$ in KNN.

Answer Guide

Bias is error from overly simple assumptions. Variance is error from sensitivity to training data changes. Good models balance both.

Case	Bias/Variance interpretation
Model too simple	High bias, low variance -> underfitting
Model too complex	Low bias, high variance -> overfitting
$k=1$ in KNN	High variance, noise-sensitive
$k=n$ in KNN	High bias, predicts majority/global average

Q2. A) Explain KNN edge cases. B) What happens when $k=1$, $k=n$, k is even in binary classification, or dataset has class imbalance?

Answer Guide

- $k=1$: nearest point controls prediction; highly noise-sensitive; overfitting/high variance.
- $k=n$: all points vote; classification predicts majority class, regression predicts global average; underfitting/high bias.
- Even k in binary classification: tie possible; use odd k or tie-breaking/distance-weighting.
- Class imbalance: majority class may dominate votes; use weighted voting, balanced sampling, or better metrics like recall/F1.

Q3. A) Explain SVM using margin and support vectors. B) What is the kernel trick and why is it useful for non-linear classification?

Answer Guide

SVM finds a decision boundary that maximizes the margin between classes. The points closest to the boundary are support vectors; they determine the boundary.

The kernel trick allows SVM to create nonlinear decision boundaries by computing similarity as if data were mapped to a higher-dimensional feature space, without explicitly computing that mapping.

Exam memory: SVM = margin-based classifier; support vectors = boundary-defining points; kernel = nonlinear separation tool.

Q4. A) Explain Apriori principle. B) If there are 10 unique items, calculate total itemsets including empty set, non-empty itemsets, and total possible association rules.

Answer Guide

Apriori principle: if an itemset is frequent, all of its subsets must also be frequent. If an itemset is infrequent, all of its supersets must be infrequent.

$n=10$

Itemsets including empty set = $2^{10} = 1024$

Non-empty itemsets = $2^{10} - 1 = 1023$

Association rules = $3^{10} - 2^{(10+1)} + 1 = 59049 - 2048 + 1 = 57002$

Practice Paper 8: Dataset Formats, Metrics, Imbalance, PCA

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain record, transaction, document, graph, and ordered data. B) Which type suits association rule mining, text classification, social network analysis, and time-series forecasting?

Answer Guide

Data format	Meaning	Suitable task
Record data	Rows and columns	General ML models
Transaction data	Basket/item data	Association rules
Document data	Text/document-term matrix	Text classification
Graph data	Nodes and edges	Social network analysis
Ordered data	Sequence/time-based records	Time-series forecasting

Mapping: association rule mining -> transaction data; text classification -> document data; social network analysis -> graph data; time-series forecasting -> ordered data.

Q2. A) Explain confusion matrix. B) Given TP=36, FP=6, FN=14, TN=44, calculate accuracy, precision, recall, F1-score, and specificity.

Answer Guide

A confusion matrix summarizes actual vs predicted classes using TP, FP, FN, and TN.

$$\text{Total} = 36 + 6 + 14 + 44 = 100$$

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total} = (36 + 44) / 100 = 0.80$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 36 / (36 + 6) = 36 / 42 = 0.857$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 36 / (36 + 14) = 36 / 50 = 0.72$$

$$\text{F1} = 2\text{PR} / (\text{P} + \text{R}) = 2 * 0.857 * 0.72 / (0.857 + 0.72) = 0.782$$

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) = 44 / (44 + 6) = 44 / 50 = 0.88$$

Q3. A) Explain why accuracy can be misleading in imbalanced classification. B) A fraud classifier predicts all 1000 transactions as non-fraud. If 980 are non-fraud and 20 are fraud, calculate accuracy and explain why model is bad.

Answer Guide

Accuracy can be misleading when one class dominates. A model can get high accuracy by always predicting the majority class while completely missing the minority class.

$$\text{Correct predictions} = 980 \text{ non-fraud}; \text{Total} = 1000$$

$$\text{Accuracy} = 980 / 1000 = 98\%$$

The model is bad because it detects zero fraud cases. Fraud recall = $0 / 20 = 0$. High accuracy hides minority-class failure.

Q4. A) Explain PCA and why dimensionality reduction is useful. B) What is explained variance in PCA?

Answer Guide

PCA transforms correlated features into new orthogonal directions called principal components. The first components capture the maximum variance in the data.

- Dimensionality reduction reduces noise, visualization difficulty, storage, and computational cost.

- It can help with high-dimensional data but may reduce interpretability.

Explained variance tells how much of the original data variation is captured by each principal component. Higher explained variance means the component preserves more information.

Practice Paper 9: Hierarchical Linkage, DBSCAN Concepts

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain single, complete, average, and Ward linkage. B) Points 1,3,8,12. Using complete linkage, show first two merges.

Answer Guide

Linkage	Definition	Shape tendency
Single	Minimum pairwise distance	Can chain/irregular
Complete	Maximum pairwise distance	Compact clusters
Average	Mean pairwise distance	Balanced
Ward	Smallest increase in SSE	Compact/spherical

Complete linkage uses maximum distance between points in two clusters.

Initial distances: $d(1,3)=2$, $d(3,8)=5$, $d(8,12)=4$, $d(1,8)=7$, $d(3,12)=9$, $d(1,12)=11$

First merge: 1 and 3 at distance 2 $\rightarrow$ cluster $\{1,3\}$.

Complete distance $\{1,3\}$ to 8 = $\max(7,5)=7$; $\{1,3\}$ to 12= $\max(11,9)=11$; $d(8,12)=4$

Second merge: 8 and 12 at distance 4 $\rightarrow$ cluster $\{8,12\}$.

Q2. A) Explain why single linkage can create chaining. B) Give an example where two distant points become part of same cluster through intermediate points.

Answer Guide

Single linkage merges clusters if any pair of points is close. Therefore, a sequence of locally close points can connect distant ends.

Example: points 1,2,3,4,10 with epsilon-like nearest gaps. 1 and 4 are not very close, but 1-2-3-4 forms a chain.

In hierarchical single linkage, long elongated clusters can form because global compactness is ignored.

Q3. A) Explain DBSCAN direct density reachable, density reachable, and density connected. B) If A is not directly close to D, but $A \rightarrow B \rightarrow C \rightarrow D$ through core points, can A and D be in same cluster? Explain.

Answer Guide

- Directly density reachable: q is within epsilon of core point p.
- Density reachable: q can be reached from p through a chain of directly density-reachable points.
- Density connected: two points are reachable from a common core/dense point or cluster region.

Yes. A and D can be in the same DBSCAN cluster if there is a chain through core points. DBSCAN does not require every pair in a cluster to be close directly.

Q4. A) Explain DBSCAN varying-density limitation. B) If one cluster is dense and another sparse, explain why one global epsilon may fail.

Answer Guide

DBSCAN uses fixed epsilon and MinPts. Dense clusters need smaller epsilon; sparse clusters need larger epsilon.

- If epsilon is small: sparse cluster points may be marked noise.
- If epsilon is large: nearby dense clusters may merge incorrectly.
- Therefore one global density threshold may not fit all regions.

Practice Paper 10: Bagging, Random Forest, AdaBoost

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain bagging with bootstrap sampling and OOB samples. B) Original dataset A,B,C,D,E. Tree samples: T1=A,B,B,C,E; T2=A,D,E,E,C; T3=B,C,C,D,D. Find OOB records for each tree.

Answer Guide

Bagging trains multiple models on bootstrap samples. Bootstrap sampling means selecting rows with replacement, so some rows repeat and some are not selected. Rows not selected for a tree are OOB for that tree and can be used for internal validation.

Tree	Bootstrap sample	OOB records
T1	A,B,B,C,E	D
T2	A,D,E,E,C	B
T3	B,C,C,D,D	A,E

Q2. A) Explain how Random Forest differs from Bagging. B) Why does Random Forest use random feature selection at each split?

Answer Guide

Bagging uses random rows with replacement and usually allows every tree to consider all features at every split. Random Forest uses random rows plus a random subset of features at each split.

Random feature selection prevents all trees from repeatedly choosing the same strongest feature. This decorrelates trees, increases diversity, and makes majority voting more effective.

Memory line: Bagging = random rows; Random Forest = random rows + random feature subset at each split.

Q3. A) Explain AdaBoost workflow. B) A weak learner has weighted error 0.25. Calculate learner weight $\alpha = 0.5 \ln((1-\text{error})/\text{error})$.

Answer Guide

- Start with equal sample weights.
- Train a weak learner.
- Compute weighted error.
- Increase weights of misclassified samples.
- Decrease relative weights of correctly classified samples.
- Combine learners by weighted voting.

error=0.25

$\alpha = 0.5 \ln((1-0.25)/0.25) = 0.5 \ln(0.75/0.25) = 0.5 \ln(3)$

$\ln(3) = 1.099 \rightarrow \alpha = 0.549$

Q4. A) Explain why boosting is sensitive to noisy labels and outliers. B) Compare Random Forest and AdaBoost in training style, voting, and noise sensitivity.

Answer Guide

Boosting increases the weight of misclassified samples. If a sample is noisy, mislabeled, or an outlier, it may be misclassified repeatedly and receive too much attention, hurting generalization.

Aspect	Random Forest	AdaBoost
Training	Parallel/independent trees	Sequential weak learners
Voting	Usually equal majority vote	Weighted vote
Main effect	Reduces variance	Often reduces bias
Noise sensitivity	Lower	Higher

Practice Paper 11: Validation, Complexity, Scalability, Outliers

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain model selection and validation. B) Differentiate training set, validation set, test set, and cross-validation.

Answer Guide

Model selection is choosing the algorithm and hyperparameters that generalize best. Validation estimates performance during model selection.

Term	Purpose
Training set	Fit model parameters
Validation set	Tune hyperparameters and compare models
Test set	Final unbiased evaluation after model selection
Cross-validation	Repeated train-validation splits to get more stable estimate

Q2. A) Explain time complexity of KNN. Why is KNN called lazy learning? B) For $n=1000$ training points and $d=20$ features, approximately how many feature comparisons are needed for one KNN prediction?

Answer Guide

KNN is lazy because it stores training data and delays computation until prediction.

Prediction complexity for one query = $O(n*d)$

$n*d = 1000*20 = 20,000$ feature comparisons approximately

If full sorting is used, additional sorting cost may appear, but distance computation is the main cost in basic KNN.

Q3. A) Compare time complexity/scalability of k-means, hierarchical clustering, DBSCAN, and BIRCH. B) Which one is most scalable and why?

Answer Guide

Algorithm	Typical time idea	Scalability
k-means	$O(t*n*k*d)$	Good
Hierarchical	$O(n^2)$ to $O(n^3)$, memory $O(n^2)$	Poor for large n
DBSCAN	$O(n \log n)$ with index, worst $O(n^2)$	Depends on indexing/dimensions
BIRCH	Approximately $O(n)$	Very good

BIRCH is usually most scalable because it summarizes data into CF-tree micro-clusters and avoids storing all pairwise distances.

Q4. A) Explain outlier sensitivity in k-means, BIRCH, hierarchical clustering, and DBSCAN. B) Which algorithm handles outliers best and why?

Answer Guide

Algorithm	Outlier behavior
k-means	Highly sensitive because centroids are means and SSE squares far distances
BIRCH	Moderately sensitive; outliers may distort CF summaries or create tiny micro-clusters
Hierarchical	Depends on linkage; single can chain, complete/Ward affected by far points

Algorithm	Outlier behavior
DBSCAN	Relatively robust; isolated points can be labelled noise

DBSCAN generally handles outliers best because it explicitly identifies noise points instead of forcing every point into a cluster.

Practice Paper 12: Pruning, Attribute Splits, Lift, k-means Empty Clusters

Attempt the questions first. Then use the answer guide to check formula selection, computation steps, and interpretation.

Q1. A) Explain decision tree pruning and why it reduces overfitting. B) Differentiate pre-pruning, post-pruning, and rule pruning.

Answer Guide

Pruning reduces model complexity by stopping or removing unnecessary splits/conditions. It reduces overfitting and improves generalization.

Type	Meaning
Pre-pruning	Stop tree/rule growth early using max depth, min samples, min gain, or validation checks
Post-pruning	Grow full tree first, then remove branches that do not improve validation/generalization
Rule pruning	Remove unnecessary conditions from IF-THEN rules

Q2. A) Explain how decision trees split nominal, ordinal, and continuous attributes. B) Give one example split for each type.

Answer Guide

Attribute type	How split works	Example
Nominal	Split by category or category groups	city in {Delhi,Mumbai}?
Ordinal	Respect order while choosing thresholds/groups	size <= medium?
Continuous	Sort values and test threshold splits	age <= 35?

The tree chooses the split that gives maximum impurity reduction using entropy/IG, Gini reduction, or similar criteria.

Q3. A) Explain high confidence but low lift in association rules. B) In 100 transactions: 90 contain Butter, 50 contain Bread, 45 contain both. Calculate confidence and lift for Bread -> Butter. Explain whether rule is truly interesting.

Answer Guide

Confidence can be high just because the consequent item is very common. Lift checks whether the rule is stronger than random chance.

$$\text{Confidence}(\text{Bread} \rightarrow \text{Butter}) = \text{both}/\text{Bread} = 45/50 = 0.90$$

$$\text{Support}(\text{Butter}) = 90/100 = 0.90$$

$$\text{Lift} = \text{Confidence}/\text{Support}(\text{Butter}) = 0.90/0.90 = 1.00$$

Although confidence is 90%, lift is 1, so the rule is not truly interesting. Bread does not increase the chance of Butter beyond its normal frequency.

Q4. A) Explain k-means empty cluster problem. B) In k-means with k=100, only one non-empty cluster is formed. Explain at least four possible reasons and how practical implementations handle this issue.

Answer Guide

An empty cluster occurs when no points are assigned to one or more centroids during the assignment step.

- Poor centroid initialization.
- Duplicate or nearly duplicate centroids.

- k is too large relative to natural cluster structure.
- All points are identical or very concentrated.
- Implementation does not reinitialize empty clusters.
- All points become closest to one centroid.

Practical implementations handle this by reinitializing empty centroids, assigning them to farthest points, reducing k , or using k-means++ initialization.

Exam-ready: Yes, one non-empty cluster can occur in a degenerate case, but robust implementations usually prevent or repair empty clusters.